RETRIEVAL-AUGMENTED GENERATION BASED DOCUMENT QUESTION ANSWERING SYSTEM

FINAL PROJECT REPORT

Project Title: Paracetamol Document QA with RAG

Application Type: PDF Question Answering using Retrieval-Augmented Generation

Core Stack: Streamlit, LangChain, Hugging Face Embeddings, FAISS, Transformers

Prepared For: Academic project submission

Date: March 10, 2026

Abstract

This project presents a Retrieval-Augmented Generation (RAG) based question answering system designed for querying a PDF document about Paracetamol. Traditional question answering systems often hallucinate or respond from general model memory without grounding the answer in the source material. To address this, the proposed system combines semantic retrieval and local language generation so that answers are produced only from the document context selected at runtime.

The application accepts a PDF file through a Streamlit interface, extracts text using a PDF loader, splits the content into overlapping chunks, transforms the chunks into embeddings using `sentence-transformers/all-MiniLM-L6-v2`, and stores the embeddings in a FAISS vector index. When the user enters a question, the system retrieves the top relevant chunks and passes them to a LangChain `RetrievalQA` chain backed by a local sequence-to-sequence model, `google/flan-t5-base`, to generate a grounded answer.

The implemented system supports configurable chunk size, chunk overlap, and retrieval depth, and also displays the retrieved chunks, source chunks, and similarity scores for transparency. The result is a compact, explainable, and domain-focused document question answering pipeline suitable for educational and informational use.

Introduction

Large language models are effective at generating natural language responses, but their answers are not always anchored to a trusted source. In document-centric tasks, especially when users need precise information from a particular file, retrieval becomes necessary. Retrieval-Augmented Generation addresses this problem by first finding relevant context from the source material and then generating an answer constrained to that context.

In this project, the target document is a PDF related to Paracetamol. The goal is to allow a user to ask natural language questions such as the uses, dosage notes, warnings, or other details described in the document, and receive answers derived from the uploaded file rather than from unrestricted model memory.

> The implemented system is intentionally grounded: if the answer is not present in the retrieved context, the prompt instructs the model to respond that the answer is not known from the provided document.

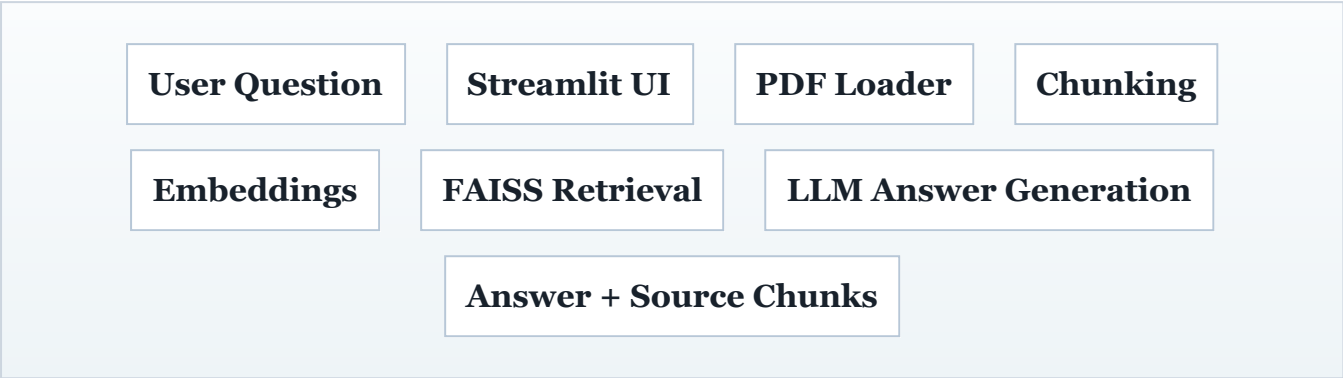
Problem Statement

Users often need a simple way to query long PDF documents without reading every page manually. Keyword search is limited because it does not capture semantic similarity and may miss relevant information expressed in different words. A domain-oriented question answering system is therefore needed to retrieve semantically relevant content and produce readable answers with supporting context.

Objectives

- Build a PDF question answering application using a standard RAG pipeline.
- Enable semantic retrieval over Paracetamol document content.
- Generate answers using only retrieved context from the selected PDF.
- Expose retrieved chunks and similarity scores for answer transparency.
- Provide an easy-to-use web interface through Streamlit.

System Architecture



Technology Stack

Component	Technology Used	Purpose
Frontend	Streamlit	Interactive web interface for upload, question input, and results display
Document Loading	LangChain Community `PyPDFLoader`	Extracts text content page by page from PDF documents
Chunking	`RecursiveCharacterTextSplitter`	Splits document text into overlapping semantic segments
Embeddings	`sentence-transformers/all-MiniLM-L6-v2`	Converts document chunks and queries into dense vector representations
Vector Store	FAISS	Stores and searches document embeddings efficiently
Generation Model	`google/flan-t5-base`	Produces final natural language answers from retrieved context
Orchestration	LangChain	Connects retriever, prompt, and local language model into one QA chain

Project Structure

File	Role in the Project
`app.py`	Streamlit interface, configuration controls, question handling, and output rendering
`rag_pipeline.py`	Core RAG logic including loading, chunking, embedding, indexing, retrieval, and answer generation
`utils.py`	Helper functions for file handling, chunk formatting, document identity, and duplicate removal

File	Role in the Project
`data/paracetamol.pdf`	Default knowledge source used when no new PDF is uploaded

DEFAULT RETRIEVAL SETUP

Chunk Size: 600

Chunk Overlap: 80

Top-k Retrieval: 3 chunks

MODEL SETUP

Embedding Model: all-MiniLM-L6-v2

Generator: flan-t5-base

Execution: Local CPU-based inference

Methodology and Implementation

1. PDF Ingestion

The system accepts either an uploaded PDF or a default file located at `data/paracetamol.pdf`. The PDF is loaded through `PyPDFLoader`, which creates page-level documents. The total page count is stored and later displayed in the interface as part of ingestion metrics.

2. Chunking Strategy

After loading, the text is split using `RecursiveCharacterTextSplitter` with a default chunk size of 600 characters and overlap of 80 characters. The splitter prioritizes separators in the order of paragraph breaks, line breaks, sentence endings, spaces, and finally character-level fallback. This preserves meaning while limiting context loss at chunk boundaries.

The chosen chunking values balance retrieval accuracy and contextual completeness. Larger chunks preserve enough surrounding medical explanation, while overlap ensures that boundary facts are not dropped when text spans adjacent chunks.

3. Embedding Generation

Each chunk is converted into a dense vector representation using the Hugging Face embedding model `sentence-transformers/all-MiniLM-L6-v2`. Query text is embedded in the same vector space, allowing semantic comparison between user questions and document content. Embedding normalization is enabled so similarity comparisons remain stable during retrieval.

4. Vector Indexing with FAISS

The embedded chunks are indexed using FAISS. A new FAISS index is built whenever the active document or retrieval configuration changes. This index acts as the retriever backend and returns the top-k most relevant chunks for each user question.

5. Answer Generation

For answer generation, the project defines a custom LangChain-compatible class, `LocalSeq2SeqLLM`, which wraps a local Transformers sequence-to-sequence model. The chosen model is `google/flan-t5-base`. The QA chain uses a prompt that explicitly restricts the model to the retrieved context and instructs it to admit when the answer is absent from the document.

6. Retrieval and Transparency

The pipeline performs a direct similarity search against FAISS and converts raw distances into a readable similarity-style score using `1 / (1 + score)`. The interface then displays the retrieved chunks, their page metadata, and the source chunks returned by the QA chain. This makes the system more interpretable than a plain black-box answer generator.

User Interface Features

- PDF uploader for custom documents.
- Fallback to a default Paracetamol PDF.

- Sidebar controls for embedding model, generation model, chunk size, overlap, and top-k.
- Question input and single-click answer generation.
- Metrics display for page count, chunk count, and retrieval depth.
- Separate sections for final answer, retrieved chunks, and source chunks used for answering.

Key Source Code Highlights

Streamlit Entry Point: `app.py`

```
@st.cache_resource(show_spinner=False)
def get_pipeline(
    chunk_size: int,
    chunk_overlap: int,
    top_k: int,
    embedding_model_name: str,
    generation_model_name: str,
) -> RAGPipeline:
    return RAGPipeline(
        chunk_size=chunk_size,
        chunk_overlap=chunk_overlap,
        top_k=top_k,
        embedding_model_name=embedding_model_name,
        generation_model_name=generation_model_name,
    )
```

Chunking and FAISS Indexing: `rag_pipeline.py`

```
splitter = RecursiveCharacterTextSplitter(
    chunk_size=self.chunk_size,
    chunk_overlap=self.chunk_overlap,
    separators=["\n\n", "\n", ". ", " ", ""],
)
chunks = splitter.split_documents(documents)
self.vectorstore = FAISS.from_documents(chunks, self.embeddings)
```

Answer Generation with RetrievalQA

```
self.qa_chain = RetrievalQA.from_chain_type(
    llm=self.llm,
    chain_type="stuff",
    retriever=retriever,
    return_source_documents=True,
    chain_type_kwargs={"prompt": self._build_prompt()},
)
```

Working Flow of the System

1. The user uploads a PDF or uses the default Paracetamol document.
2. The PDF is parsed into page-level text documents.
3. The text is split into overlapping chunks.
4. All chunks are embedded and stored in a FAISS vector index.
5. The user asks a natural language question.
6. The question is embedded and matched against the vector index.
7. The top relevant chunks are retrieved.
8. The prompt combines those chunks with the user question.
9. The local language model generates a grounded answer.
10. The system displays the answer and the supporting chunks.

Key Implementation Decisions

Decision	Rationale
Use of local generation model	Reduces dependence on external APIs and keeps the project self-contained.
Use of MiniLM embeddings	Provides strong semantic retrieval quality with a lightweight model size.
Use of FAISS	Enables fast similarity search over document chunks.
Grounded prompt design	Helps minimize hallucinated answers outside the given PDF context.
Cached pipeline object in Streamlit	Avoids unnecessary reinitialization and repeated indexing on every question.

Observations and Results

The completed system demonstrates that a compact RAG pipeline can answer domain-specific questions from a PDF more reliably than unrestricted generation alone. Because answers are formed after semantic retrieval, the output remains tied to the content of the chosen document. The display of retrieved chunks and source chunks also improves user trust by showing the basis of the answer.

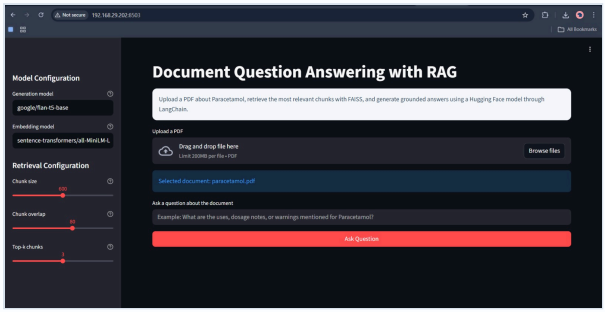
The application is especially suitable for short to medium-length reference documents where users need fast access to factual content such as usage instructions, dosage notes, warnings, or definitions. The configurable retrieval settings also make the project useful for experimentation and learning.

Output Screenshots

The following screenshots were captured from the running Streamlit application. They confirm that the interface, question submission flow, and retrieval output are functioning in the implemented system.

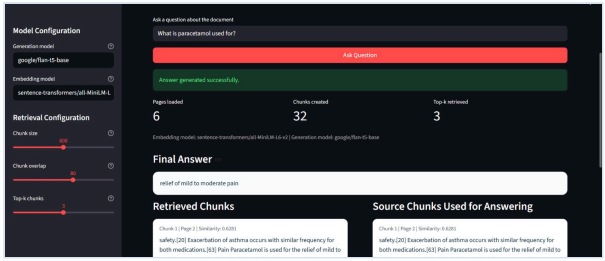
Screenshot 1: Home Interface

Initial application screen with the title, uploader, question field, and sidebar configuration controls.



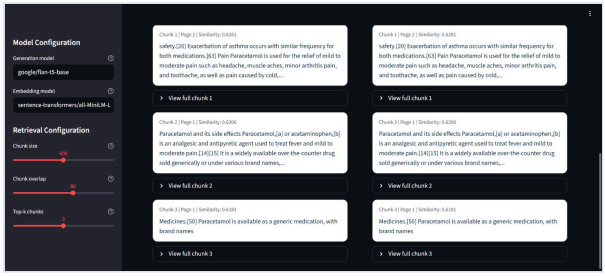
Screenshot 2: Question Submission View

Application state after entering a query for the Paracetamol document question answering workflow.



Screenshot 3: Retrieved Chunks

Retrieved chunk output with evidence cards, page references, and similarity scores displayed to the user.



Suggested Example Queries for Screenshot Capture

- `What is paracetamol used for?`
- `What dosage information is mentioned for paracetamol?`
- `What warnings or precautions are described in the document?`
- `What side effects are mentioned in the PDF?`

Limitations

- The quality of answers depends on the quality and extractability of text from the PDF.
- Very complex layouts, tables, or scanned pages may reduce extraction quality.
- The final answer quality is bounded by the capability of the local generator model.
- No quantitative evaluation benchmark is included in the current version.

Conclusion

This project successfully implements a Retrieval-Augmented Generation based PDF question answering system for Paracetamol-related content. The application combines document loading, recursive chunking, semantic embeddings, FAISS-based retrieval, and local language generation into one usable interface. The design meets the practical objective of allowing a user to ask questions in natural language and receive grounded answers supported by retrieved source text.

Beyond the specific Paracetamol document, the architecture is reusable for many other domain-focused document QA tasks. This makes the project a strong demonstration of how RAG systems can improve factual reliability and usability in document intelligence workflows.

Future Scope

- Add persistent vector storage so the index can be reused across sessions.
- Support multiple uploaded documents and document selection.
- Introduce evaluation sets for answer relevance and grounding accuracy.
- Improve handling of scanned PDFs through OCR.
- Add citation-style sentence references in the generated answer.

References

- LangChain documentation for retrieval chains and document loaders.
- Hugging Face model documentation for sentence-transformers and FLAN-T5.
- FAISS documentation for vector similarity search.
- Streamlit documentation for Python-based data applications.
- Project source files: `app.py`, `rag_pipeline.py`, and `utils.py`.

Report content aligned to the implementation present in the project workspace on March 10, 2026. No unsupported quantitative claims were added.